

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO: 40. Draw Rectangle and Extract Object Using OpenCV

Aim

To draw a rectangular bounding box around an object and extract the object from BEN 10.png using OpenCV.

Algorithm

1. Import OpenCV.
2. Read the image BEN 10.png.
3. Convert the image to grayscale.
4. Apply thresholding.
5. Find contours in the image.
6. Find the largest object.
7. Draw a rectangle around the object.
8. Extract the object using the rectangle coordinates.
9. Display and save the result.

Code:

```
EXP-40.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
cv2.CHAIN_APPROX_SIMPLE
)
if len(contours) == 0:
    print("No object found!")
    exit()

# Find largest contour
largest = max(contours, key=cv2.contourArea)

# Get bounding rectangle
x, y, w, h = cv2.boundingRect(largest)

# Draw rectangle
result = img.copy()

cv2.rectangle(
    result,
    (x, y),
    (x + w, y + h),
    (0, 255, 0),
    2
)

# Extract object
object_image = img[y:y+h, x:x+w]

# Display results
cv2.imshow("Object with Rectangle", result)
cv2.imshow("Extracted Object", object_image)

# Save results
cv2.imwrite("EXP40_rectangle.png", result)
cv2.imwrite("EXP40_extracted_object.png", object_image)

print("Rectangle drawn successfully!")
print("Object extracted successfully!")

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT:

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-38.py
Traceback (most recent call last):
  File "C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-38.py", line 15, in <module>
    face_cascade = cv2.CascadeClassifier(
AttributeError: module 'cv2' has no attribute 'CascadeClassifier'
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-38.py
Face/Object detected successfully!
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-39.py
Vehicle detection complete
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-39.py
Vehicle detection complete
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-40.py
Rectangle drawn successfully
Object extracted successfully
```



Result

The object in BEN 10.png is successfully enclosed within a rectangular bounding box and extracted using OpenCV.